

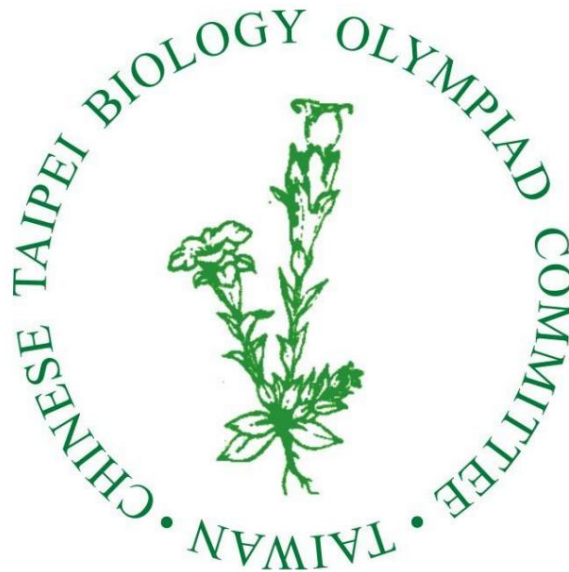
2021 Republic of China Biology Olympiad

National Team Selection Camp

Theoretical Assessment Test Questions

Part One

Student Code: Preparatory Paper



Precautions:

1. This section of the test consists of 9 pages (including the cover) and is worth 100 points.
2. Please check for missing pages. If any pages are missing, please inform the test administrator immediately.
3. When answering the questions, please make sure that the cover of the exam paper matches your exam number.
4. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.
5. You may write your answers on the back of the questions, but please indicate the question number.

Part 1: (100 points)

I. International Competition Type

Problems y Problems (1-10)

International competition format: Unequal scoring. Each question presents four statements. Judge whether the four statements are correct and fill in T (correct) or F (incorrect) in the corresponding column of the table below. Correct answers for the first, second, third, and fourth statements are worth 0.5, 1, 2.5, and 5 points respectively. (5 points per question, 50 points in total)

Multiple choice question number	(A)	(B)	(C)	(D)	Fraction
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

1. Chloroplasts are the most important and ubiquitous plastids in plant cells, and they are organelles that perform photosynthesis.

(A) A chloroplast is an inwardly folded, double-membrane organelle.

(B) Archaea are considered the ancestors of chloroplasts.

(C) In C₄ type plants, the vascular bundles are surrounded by bundle sheaths, and the cells inside also contain chloroplasts, which are a secondary component of the light response.

Position

(D) From the outside in, it can be divided into six functional regions: the outer chloroplast membrane, the intermembranous space of the chloroplast, the inner chloroplast membrane, the stroma, the chloroplast thylakoid membrane, and the thylakoid cavity of the chloroplast.

2. In biology, the genome refers to the DNA contained in an organism (some viruses are...).

All the genetic information contained in RNA.

(A) Includes genes and non-coding DNA. (B)

Generally speaking, the larger the genome size, the more genes are represented. (C)

Genome size refers to the total number of DNA base pairs in an organism's haploid genome. (D) In eukaryotes,

the majority (>80%) of the genome is non-repetitive DNA, meaning that most of it contains functional coding DNA.

DNA

3. Programmed cell death (PCD) is a collective suicide of cells in multicellular organisms according to a predetermined program. It includes two types: apoptosis and autophagy. When PCD occurs, cells exhibit the following characteristics.

(A) The cell will rupture and release its contents.
 (B) The nuclear lamina will break and disappear. (C) Bubbling or blistering phenomena will appear on the cell surface, forming spherical protrusions. (D) Nuclear DNA will break at the nucleosome junctions to form nucleosome fragments.

4. The third type of fibrous structure in the cytoskeleton is called intermediate filaments, also known as intermediate threads, which are hollow. It has a bony structure with tension and shear resistance.

(A) Smallest in diameter (B) Least stable in the cytoskeleton
 (C) Primarily composed of γ -helices and (D) The most complex component of the cytoskeleton.

5. Peptide synthesis is the process of synthesizing peptides in organic chemistry. Peptides are organic compounds composed of multiple amino acids linked together by peptide bonds. In biology, the process of synthesizing long peptides (proteins) is called protein biosynthesis.

(A) The larger the single peptide molecule, the slower its synthesis rate. (B) The more complex the single peptide molecule, the slower its synthesis rate. (C) All single peptide molecules have roughly similar synthesis rates. (D) The synthesis rates of prokaryotes and eukaryotes are similar.

6. DNA methylation is a form of chemical modification of DNA. During methylation, methyl groups are... (methyl-) is added to the DNA molecule.

(A) In human genes, methylation mainly occurs at CpG sites in the gene sequence. (B) It can alter genetic expression without changing the DNA sequence. (C) Tumor cells exhibit methylation imbalance, meaning that methylation occurs at promoter sites throughout the entire gene. (D) DNA demethylation in prokaryotic cells can act as a primitive immune system, protecting cells from bacteriophage infection.
 dye

7. Cyclin E forms a complex with cyclin kinase Cdk2, regulating the cell cycle transition from the first interphase (Gap1 phase) to the synthesis phase. Please determine whether the following statements are correct or incorrect.

(A) Cyclin E content is highest during the synthesis phase. (B) Cyclin E content is highest during the first interphase. (C) Cyclin E is regulated by cell cycle changes in terms of production and degradation. (D) The activity of the cyclin E/Cdk2 complex is highest during the second interphase (Gap 2 phase).

8. The following statements explain why red algae appear red. Please determine whether each statement is true or false.

(A) They have pigments that reflect and emit red light.
 (B) They use red light for photosynthesis.
 (C) They lack chlorophyll
 (D) They contain phycoerythrin.

9. During the breeding season, female silkworm moths secrete a chemical substance to attract male silkworm moths. This chemical substance is associated with fungi.

What substance is similar to that secreted during sexual reproduction? Please determine whether the following statements are correct or incorrect.

- (A) Chitin (B)
- Lysergic acid (C)
- Aflatoxin (D) Pheromones

10.

Before 1960, only two types of viral hepatitis were known: one transmitted through contaminated water or food, and the other through blood or bodily fluids. The former, now known as hepatitis A, causes illness lasting from weeks to months and usually has no serious consequences. The latter, now known as hepatitis B (including C), leads to chronic hepatitis, cirrhosis, and liver cancer, causing death; complications often appear years after infection. In the 1970s, researchers discovered that some patients were not infected with hepatitis A (HAV) or hepatitis B (HBV), but rather with an unknown virus. More than a decade of research failed to isolate this virus. It wasn't until Houghton extracted DNA fragments from the blood of infected chimpanzees that the virus first replicated, becoming known as hepatitis C virus (HCV). In the late 1990s, Rice genetically engineered a variant of the virus's RNA and injected it into the livers of chimpanzees, resulting in changes similar to chronic disease, confirming that HCV could cause infection and liver disease. The genetic differences in the viral samples indicate that a certain block of the HCV gene may be key to viral replication.

- (A) HAV is a small icosahedral ssRNA virus, while HBV and HCV are medium-sized spherical viruses.
(spherical) dsDNA virus
- (B) The researchers who discovered human papillomavirus (HPV), human immunodeficiency virus (HIV), hepatitis B virus (HBV), and hepatitis C virus (HCV) that cause cervical cancer were all awarded the Nobel Prize in Physiology or Medicine.
- (C) Following the discovery of HCV, diagnostic reagents for blood testing developed rapidly. Currently, tests include HCV surface antigens and surface antigens.
Antibody, e antigen, e antibody, core antigen, core antibody
- (D) To prevent the spread of hepatitis through blood, blood transfusions must be screened for hepatitis viruses beforehand. However, many people still develop chronic hepatitis after receiving screened blood, and for decades the cause remained unknown in medical history.

Questions (11-12) are

international competition-style questions: Unequal scoring. Each question has six statements; correct answers to the first, second, third, fourth, fifth, and sixth statements are worth 0.5, 1, 2, 3.5, 6, and 10 points respectively. (10 points per question, 20 points in total)

Multiple choice question number	(A)	(B)	(C)	(D)	(AND)	(F) score	
11							
12							

11.

Rabies is an encephalomyelitis caused by the rabies virus. Initial symptoms include fever, sore throat, chills, malaise, loss of appetite, vomiting, difficulty breathing, cough, weakness, anxiety, and headache. After several days, excitement and fear develop. It then progresses to paralysis, difficulty swallowing, and spasms of the throat muscles, causing hydrophobia. This is followed by mental confusion and convulsions. Without treatment, death often occurs within 2-6 days due to respiratory paralysis. The mortality rate is nearly 100%. Rabies is a zoonotic disease. After being bitten or scratched by an animal, a rabies vaccine should be administered as directed by a veterinarian. The animal should be kept under observation for 10 days. If infected with rabies, symptoms will appear within 5-8 days. Those bitten should receive five doses of rabies vaccine according to the prescribed schedule. The first dose should be injected intramuscularly into the deltoid muscle immediately on the day of the bite, along with immunoglobulin injections at other sites. The remaining four doses should be administered on days 3, 7, 14, and 28. Rabies virus laboratory workers, animal disease control personnel, dog catchers, veterinarians, animal conservationists, firefighters, and patrol officers are considered high-risk groups and should proactively receive vaccination to achieve immunization. The rabies vaccine is an attenuated vaccine, made using a weakened live virus. This method of production has been used since 1882 when microbiologist Pasteur successfully cured rabies patients. This vaccine is often made from sheep brain, which has a 1/200-1/1600 chance of causing central or peripheral nervous system complications. Vaccines made from newborn mouse brain are considered safer. To prevent pets from transmitting rabies to humans, cats and dogs over three months of age should be vaccinated against rabies, and then annually thereafter. New tags or microchips not only confirm vaccination but also help in finding lost pets.

Determine whether the following statements are true or false, and fill in **T** (True) or **F** (False) in the corresponding column of the table above.

(A) If immunoglobulin preparations are mixed with rabies vaccine and injected together, the synergistic effect of the two can further enhance the host's immune response.

Effect

(B) In addition to bites from infected animals, contact with the saliva, respiratory secretions, or digestive secretions of rabies patients, or through the respiratory tract, can also lead to rabies infection.

Organ transplantation can also lead to rabies infection.

(C) The purpose of vaccinating humans or pets against rabies is to induce active immunity, while the purpose of injecting immunoglobulin is to induce passive immunity in the bitten person.

(D) Rabies virus is a double-stranded RNA mantle virus. Vaccination can cause side effects, the virus may mutate or become reactivated, and there is a concern about transmitting scrabs. Therefore, it has never been universally administered in

Taiwan. (E) Vaccination should ideally be given before infection, and multiple booster shots are needed to form memory cells. Rabies vaccine is only administered after being bitten by an infected animal because a bite or scratch will inevitably trigger tetanus toxoid and immunoglobulin, producing an acute response sufficient to cope with the emergency.

(F) Animals infected with rabies may exhibit aggressive behavior because the virus replicates in the dorsal root ganglia and ascends along the spinal cord to the brain, where it proliferates in the cortical layer of the limbic system. This area governs emotions and hunger, leading to neuronal dysfunction. The animal is unable to control its low serotonin levels, resulting in anxiety, agitation, irritability, violence, and loss of attention.

12.

Protein tyrosine kinase (PTK) contains a dimeric structure and can act as a receptor for many different growth factors to regulate cell growth, differentiation, and signal transduction.

When one monomer of tyrosine kinase (TK) binds to a growth factor on the cell surface, TK becomes activated and phosphorylates the tyrosine residue of the other monomer. The phosphorylated tyrosine can then trigger signal transduction from factors such as GTPase-activating protein (GAP). Growth factor receptor-binding protein-2 (Grb2) contains an SH2 region, which can serve as the starting point for signal transduction pathways. The SH2 region contains kinases such as ras GTPase-activating protein (GAP). Growth factors bind to PTK, phosphorylating the tyrosine residue on the PTK. After PTK binds to the SH2 region, the adjacent SH3 regions connect with Sos. Sos converts Ras from the inactive GDP type to the activated GTP type. Ras is a G protein that binds GTP, produced by *the ras* gene, and can be found in cancers such as adenocarcinomas and pancreatic cancer. Activated Ras then activates the Raf protein, which in turn phosphorylates mitogen-activated protein kinase (MAPK). Phosphorylated MAPK transmits the signal into the nucleus, activating transcription factors such as *Fos*, *jun*, and *Myc*, leading to gene expression. Normal cells can return to a resting state via phosphorylated protein phosphatases, and then GAP deactivates the activated Ras-GTP to Ras-GDP, preventing the continuous generation of new cells. Mutant Ras has a higher Ras-GTP level than normal Ras. Although it can still bind to GAP, it will not activate GAPase. Mutant Ras will not be activated to Ras-GDP, and cancer cells will continue to proliferate.

Determine whether the following statements are true or false, and fill in T (True) or F (False) in the corresponding column of the table above.

- (A) The signal transduction pathway is very important. The 1994 Nobel Prize in Physiology or Medicine was awarded to two American scholars for their discovery of G proteins and their role in signal transduction in cells.
- (B) Cell membrane receptors are activated by binding to specific ligands, which phosphorylate a series of inactive protein kinases, thus converting extracellular messages into intracellular messages that are transmitted to the cell nucleus.
- (C) The final step in each protein kinase activation pathway is the activation of a dormant transcription factor in the cell nucleus, which then binds to a sequence on the promoter of a specific gene.
- (D) Problems with the cell signaling pathways that control cell growth and division can lead to cancer, immunodeficiency, and other health issues.
Besides diseases such as diabetes mellitus and autoimmune disorders, it could also be a cause of diabetes.
- (E) A cell only performs the same message transmission at a time; if different message transmissions occur simultaneously, it will cause activation.
Overlapping pathways and disruptions in cellular signaling activate different transcription factors.
- (F) Cancer chemotherapy drugs must target the synthesis and translation of cancer cell DNA, or interfere with the formation of its spindles. Effective and specific anticancer drugs cannot be designed targeting PTK, which regulates cell growth and differentiation.

II. Short Answer and Collaboration Questions y Questions

(13-15) Short Answer and Collaboration Questions: Points will be awarded based on the description of the question. (Total 30 points)

13. The Winogradsky column is an experimental apparatus invented in the 1880s by the Russian microbiologist Sergei Winogradsky (1856-1953) to simulate a microbial ecosystem. He used a pond...

Add fibrous material (shredded paper can be used as a carbon source), eggshells (containing calcium carbonate), and egg yolks (as a base) to the sludge at the bottom.

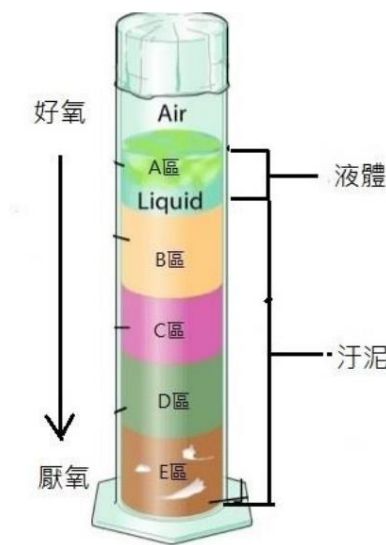
(As a source of sulfur), it is then placed in a transparent glass tube, with pool water poured into the top, the water level must be higher than the sludge above the tube.

There is a layer of liquid water, and then the unsealed tubing is exposed to sunlight for a period of time (usually several weeks). This tubing will then...

A gradient distribution gradually forms at the bottom, transitioning from aerobic to anaerobic. The hydrogen sulfide (H_2S) released after protein decomposition by bacteria can also provide raw materials for photosynthesis in some photosynthetic bacteria. Various photosynthetic bacteria are influenced by the oxygen and hydrogen sulfide content within the tube.

This process creates colored plaque areas at different depths within the column, allowing researchers to take samples from different parts of the column for culture.

Various bacterial strains with different characteristics can be isolated for research, and the growth, decline and distribution of photosynthetic bacteria can also be observed.



Schematic diagram of Vinogradsky tubular column

13-1. What type of bacteria will appear in the green area A? (2 points) Explain its important characteristics and the reason for its presence in this area.

(3 points)

13-2. What type of bacteria will appear in the purplish-red C region? (2 points) Explain its important characteristics and the reasons why bacteria appear in this region.

(3 points)

13-3. What type of bacteria would appear in the dark brown E region? **(2 points)** Explain its important characteristics and the reason for its presence in this region.
(3 points)

14. The table below lists the habitats and two key characteristics of five microorganisms: a, b, c, d, and e.

Microbial identification, main habitat, key characteristics 1			Key Feature 2
a	ruminant digestive tract	Foreign Camp	Methane can be released during metabolism.
b	Near the underwater volcano	Self-operated	Capable of oxidizing sulfides
c	Human urinary tract	Foreign Camp	The number of genes on the genome is 2368.
d	Soils in volcanic areas	aerobic	It can produce endospores
e	farmland soil	Facultative anaerobic cell walls contain peptidoglycan components.	

Description 1. Microorganism a is an archaea of prokaryotic cells.

Description 2. Microorganism b needs light to grow.

Description 3. Microorganism c is a pathogenic DNA virus.

Description 4. Microorganism d is a Gram-positive prokaryotic bacterium.

Description 5. Microorganism e is a eukaryotic cell organism.

Please fill in T (for correct) or F (for incorrect) in the corresponding column of the table below. 1 point for each question.

Naming Number	1	2	3	4	5
Answer					

15. When the external environment for cell growth lacks growth factor stimulation or when cells are aging, they enter a phase of cell division cessation, namely quiescence and senescence. Quiescence-induced cell division cessation is reversible, while senescence-induced cell division cessation is irreversible. Density-dependent inhibition is a cell number control mechanism that regulates tissue proliferation within a certain range. It was first discovered when scientists cultured cells; when the number of dividing cells covered the surface of the culture vessel, cell division stopped. If some cells were removed, cell proliferation resumed until the surface of the culture vessel was completely covered by cells. (This question...)

(10 points)

(1) Please describe the possible molecular mechanism of density-dependent inhibition. (5 points) **(2)**

Please use your creativity to consider the potential applications of this discovery of "density-dependent inhibition" in clinical medical research. (5 points)

2021 Republic of China Biology Olympiad

National Team Selection Camp

Theoretical Assessment Test Questions

Part Two

Student Code: Preparatory Paper



Precautions:

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2. Please check for missing pages. If any pages are missing, please inform the test administrator immediately.
3. When answering the questions, please make sure that the cover of the exam paper matches your exam number.
4. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.
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Part Two: (100 points)

I. The following diagram shows the molecular structure contained in a plant. Answer the following questions based on this diagram: (20 points, 5 points each)

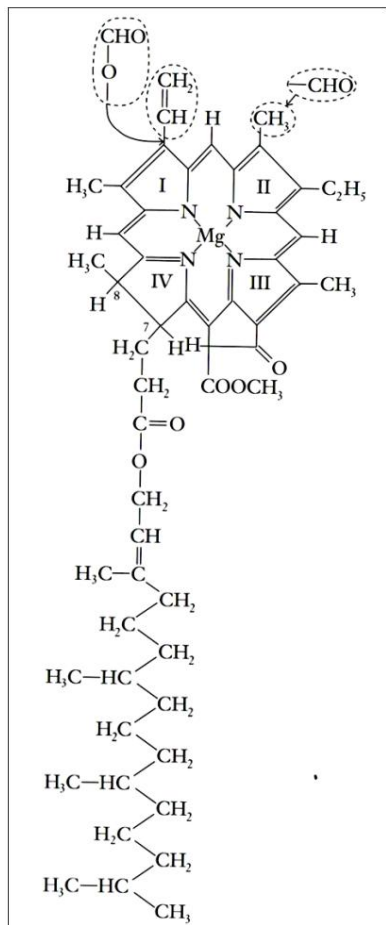


Figure A

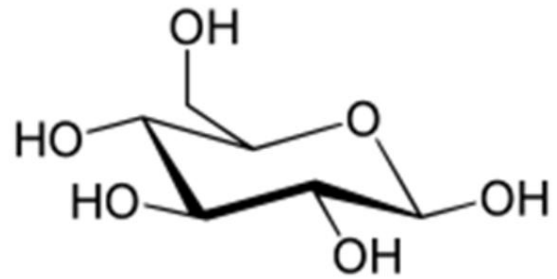


Figure B

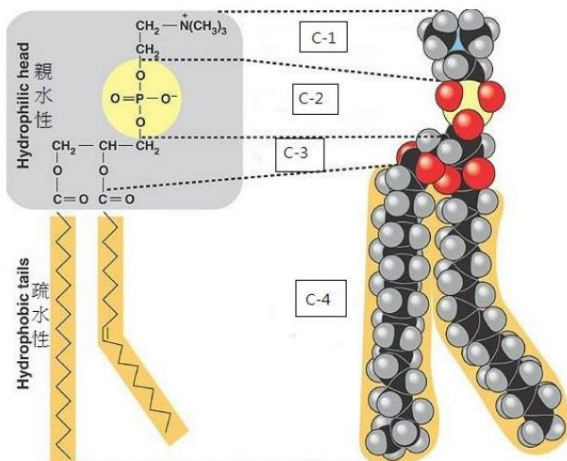


Figure C

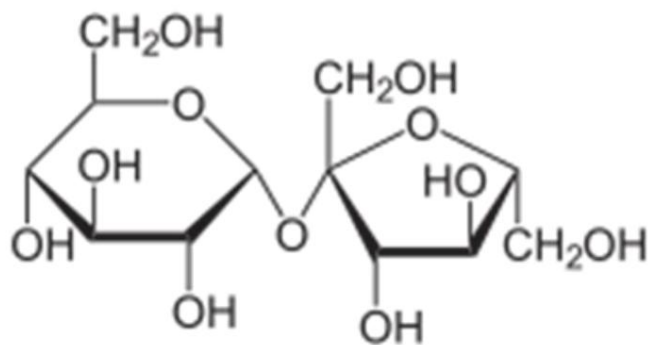


Figure D

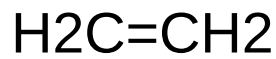


Figure E

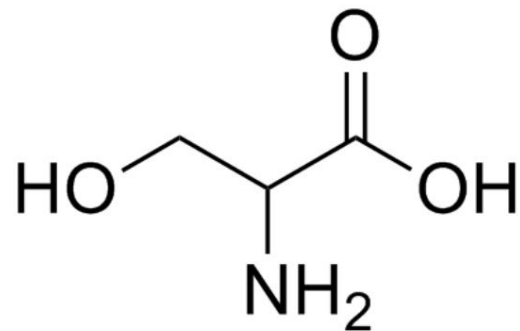


Figure F

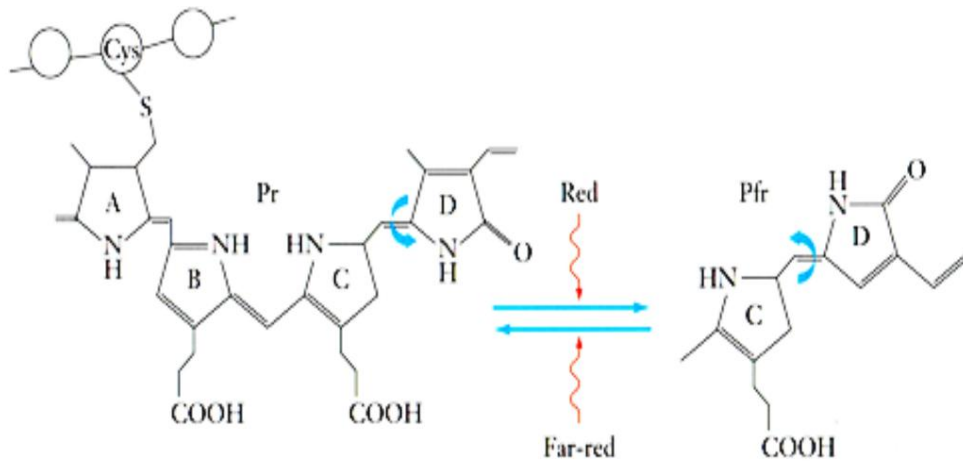


Figure G: The • symbol represents an amino acid molecule.

Answer the following questions based on the information provided: (20 points, 5 points each)

1. The molecule with photoreceptor function is shown in Figure 1. _____ With Figure _____ The physiological functions of the following are as follows: _____

9

2. The molecules commonly used in the color reaction of Benedict's solution are shown in Figure 1. _____ As shown, the gaseous hormone molecules are illustrated in the diagram. _____ As shown.

3. Why is region C-4 in Figure C hydrophobic? What is its physiological function? _____

9

4. What changes occur in the configuration of Figure G under illumination? What are its physiological functions? _____

9

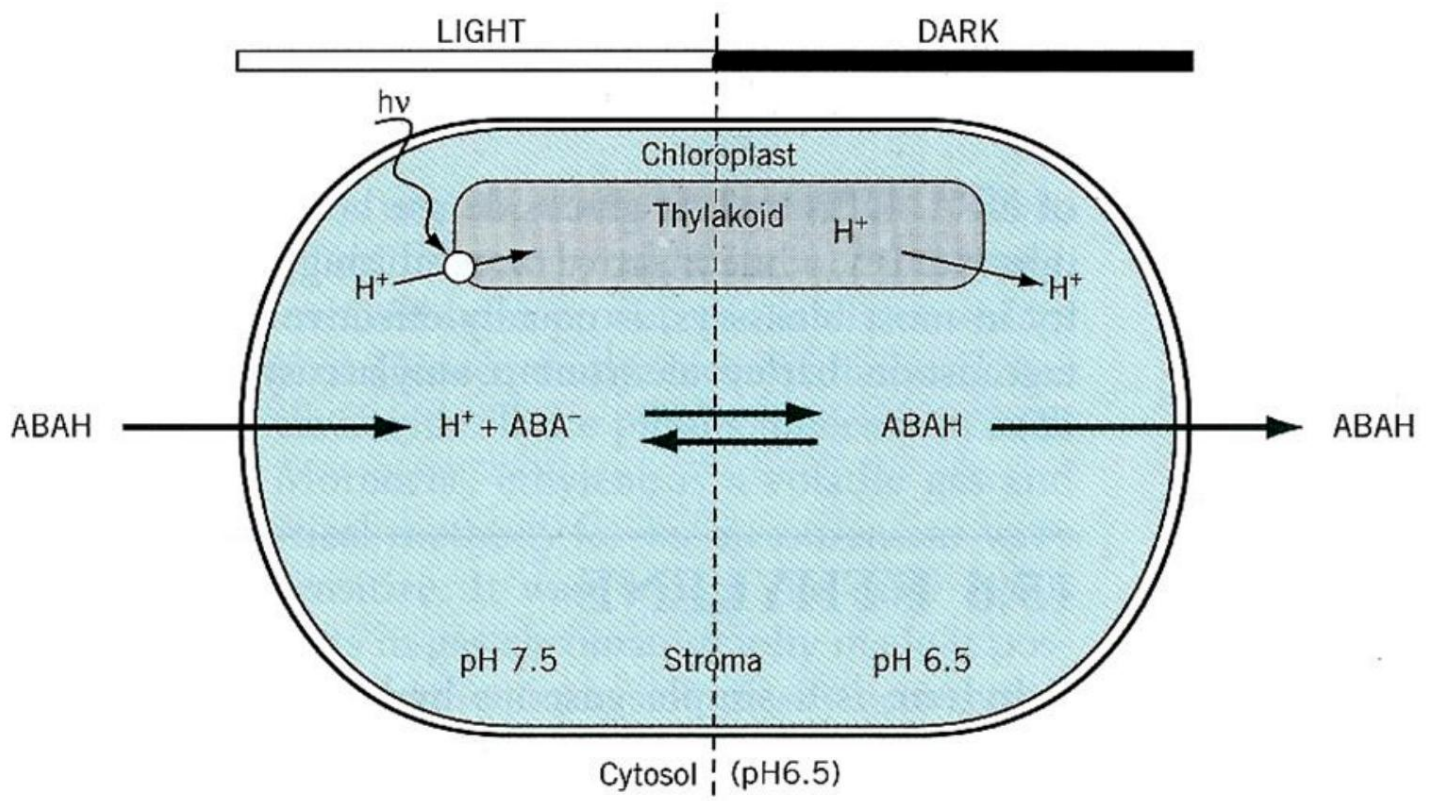
Part Two

2021 Selection Camp Theory Student Code: Preliminary Paper

II. Abscisin ABA is a key regulatory signaling molecule causing stomatal closure in plants. The image below is a schematic diagram of a plant mesophyll cell, showing abscisin ABA.

Describe the movement of ABA between the chloroplast stroma, thylakoid, and cytosol during the day and night. Answer the following questions based on this information:

(10 points, 5 points each)



1. What are the limitations on the movement of ABA between organelles?

2. What reaction produces hydrogen ions? How is the concentration gradient formed?

III. A student selected three types of permanent cross-sections of plant leaves and took 2-3 photos of each, as shown in the attached color images (Note: photos).

(The top and bottom positions are the same). Please determine its structure and answer the following questions, choosing one from the options (A–W).

Fill in the blanks with the appropriate English letters. Multiple choices are allowed. (Total 40 points)

A multi-layered epidermis	M Single-sided leaf (unifacial leaf, consisting only of mesophyll tissue on the axial or dorsal side)
B. Hypodermis	N-type leaf (equifacial leaf, with symmetrical mesophyll tissue on the axial and dorsal surfaces)
C. Palisade tissue	O keratinous hair
D. Spongy tissue	P glandular hair
E vascular bundle	Q nipple protrusion
F. Wood	R Monocotyledons
G Phloem	S Dicotyledonous plants
H-type sunken pores	T gymnosperms
I. Substomatal chamber	U xerophytes
J-axis (adaxial face)	V Aquatic plants
K-axis abaxial face	W mesophytes
L-shaped double-sided leaf (bifacial leaf, the type of mesophyll tissue differs on the axial and dorsal sides)	

1. In plant A, the black arrow in the picture refers to... _____ The red arrow refers to _____ (2 points per square)

2. In plant B, the arrow in the diagram points to... _____ (3 points)

3. In plant C, the two arrows in the diagram refer to... _____ (3 points)

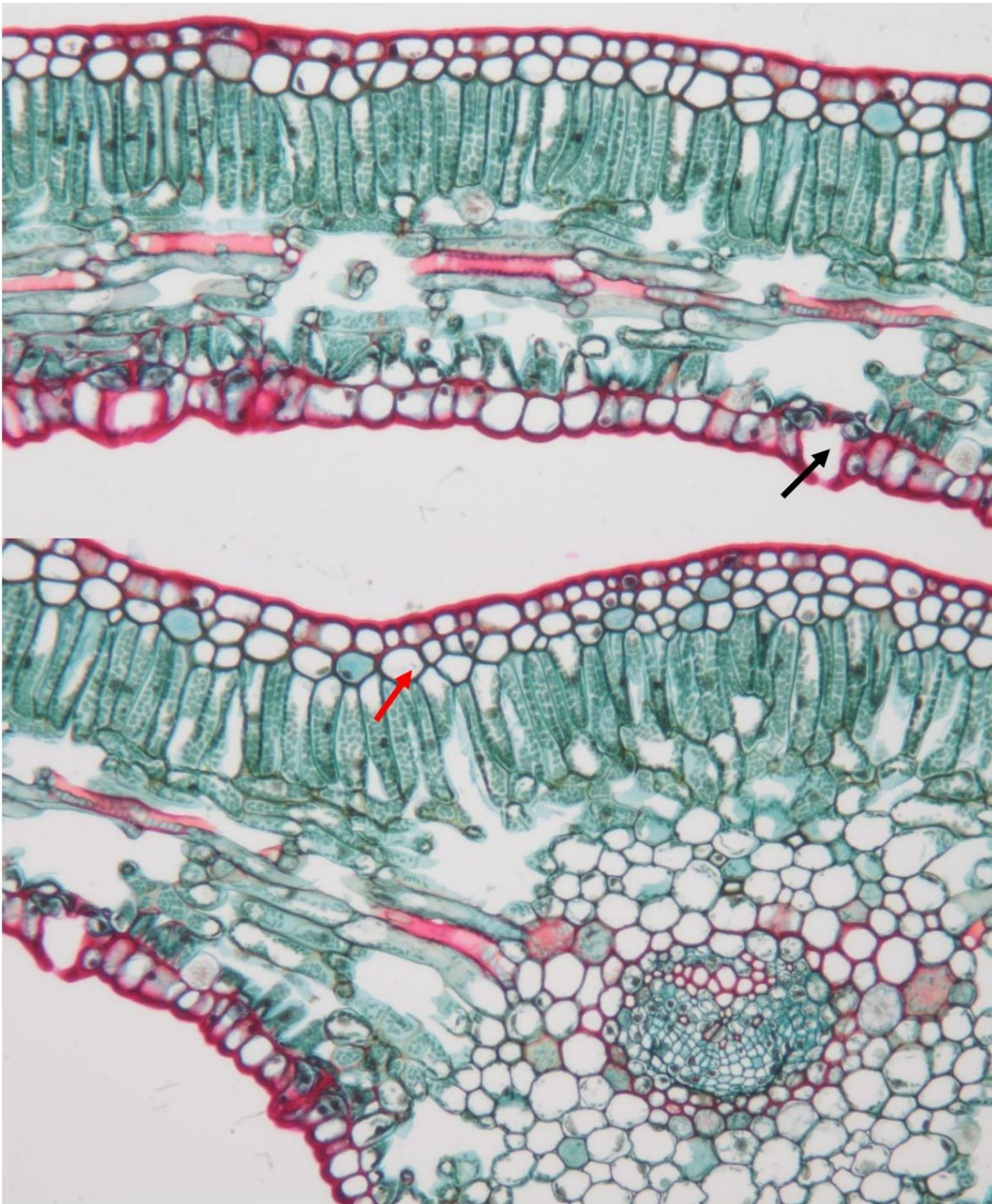
4. Based on the distribution of mesophyll tissue types, determine the leaf types of the three plants. (4 points per blank)

First: _____ ,Second: _____ C: _____

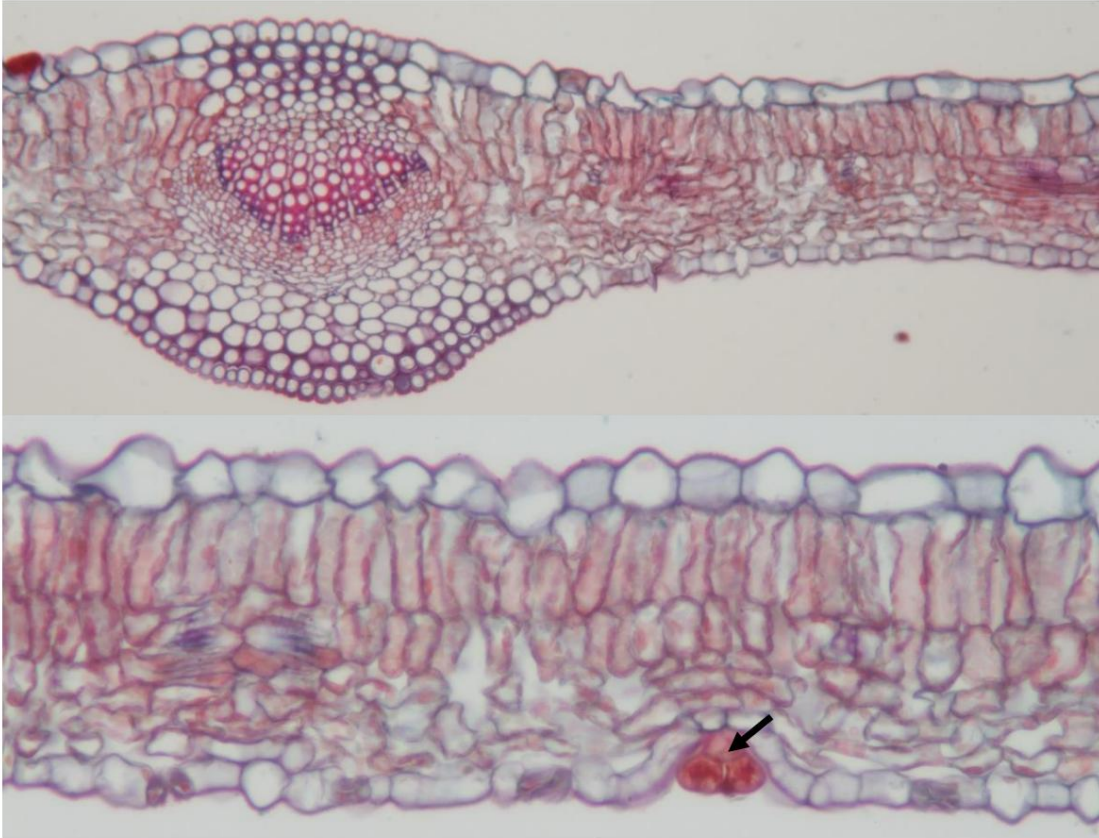
5. From a taxonomic and ecological adaptation perspective, to which type of plant do these three plants belong? (3 points each)

First: _____ ;Second: _____ C: _____

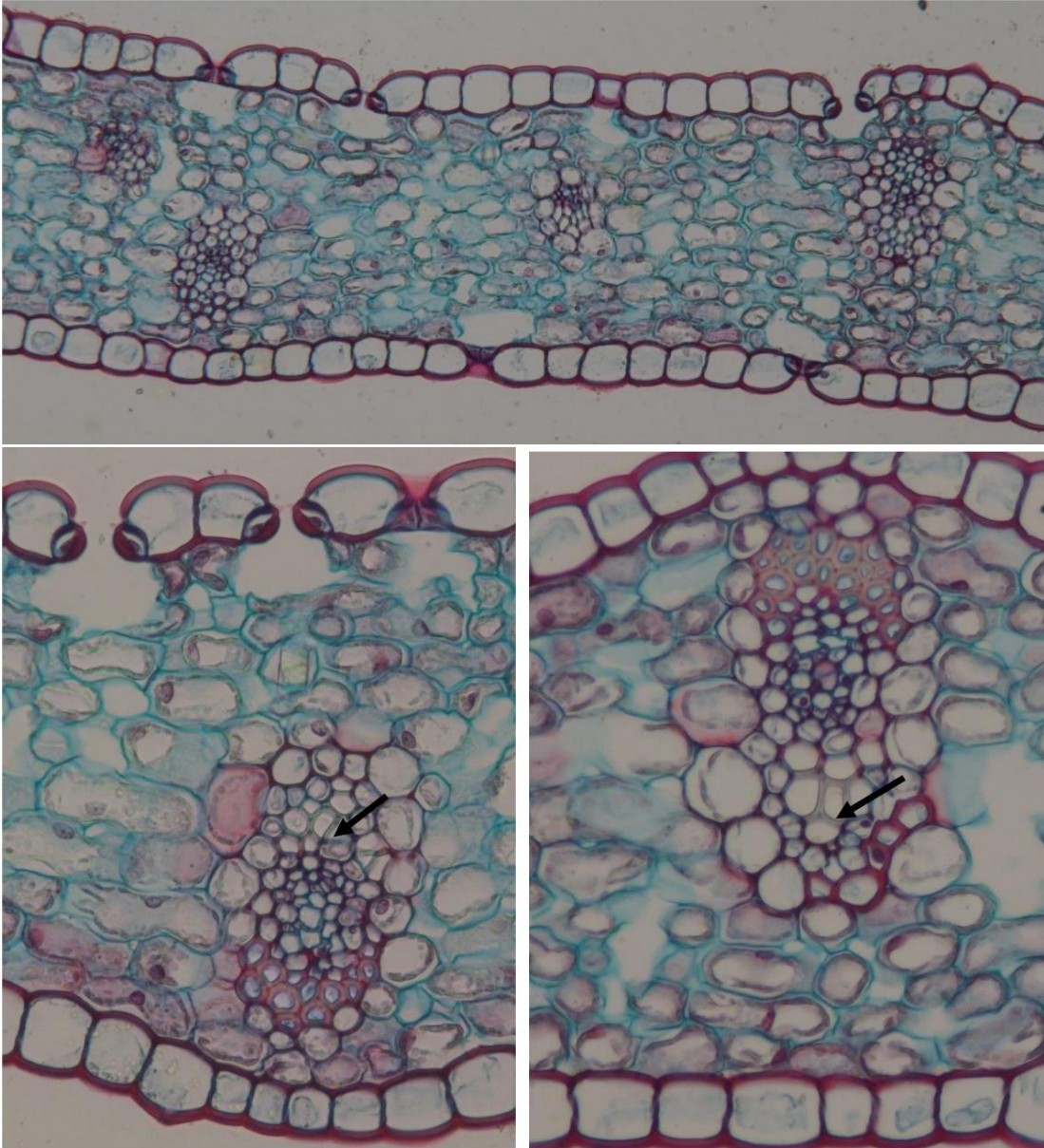
Plant A



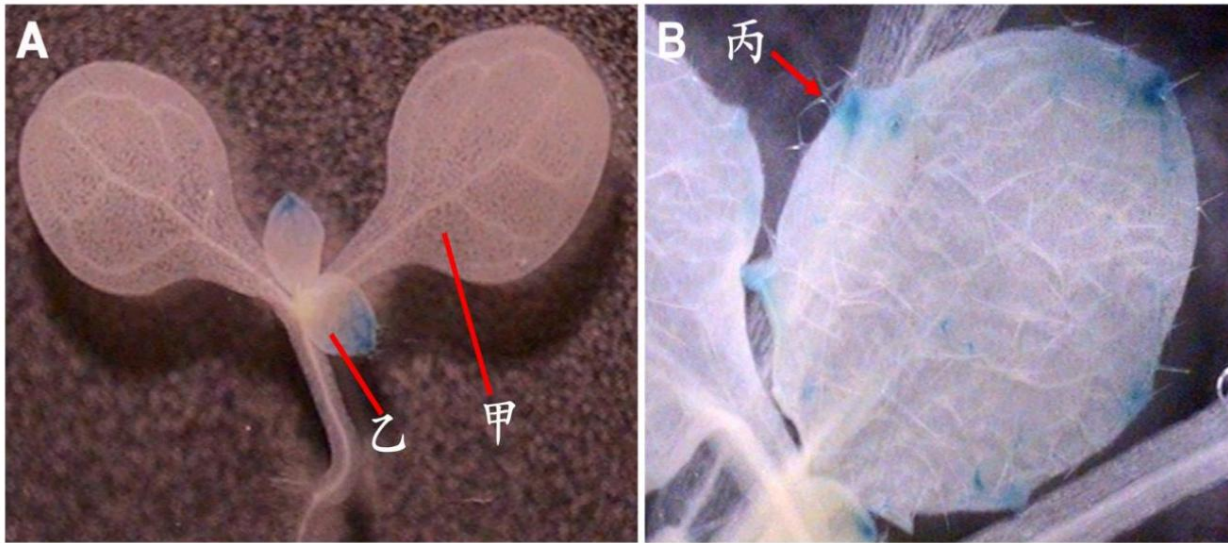
Plant B



Plant C



IV. Detecting the Expression of Plant Genes



To study the expression of gene X in a certain plant, Li Sheng established a transgenic Arabidopsis plant that uses the promoter of gene X to drive the expression of the reporter gene GUS (β -glucuronidase). The seeds of this transgenic plant were germinated and cultured under white light for 6 and 18 days. Then, using X-Gluc as a substrate, histochemical staining was performed, and the results are shown in Figure A (6-day-old seedlings) and Figure B (18-day-old seedlings) above. Based on these staining results, answer the following related questions.

1. Where is gene X expressed in this plant (please specify its location and structure)?

Six-day-old seedlings: _____ (3 points)

An 18-day-old seedling: _____ (3 points)

2. What are the differences between the surfaces of leaves A and B in Figure A? (4 points)

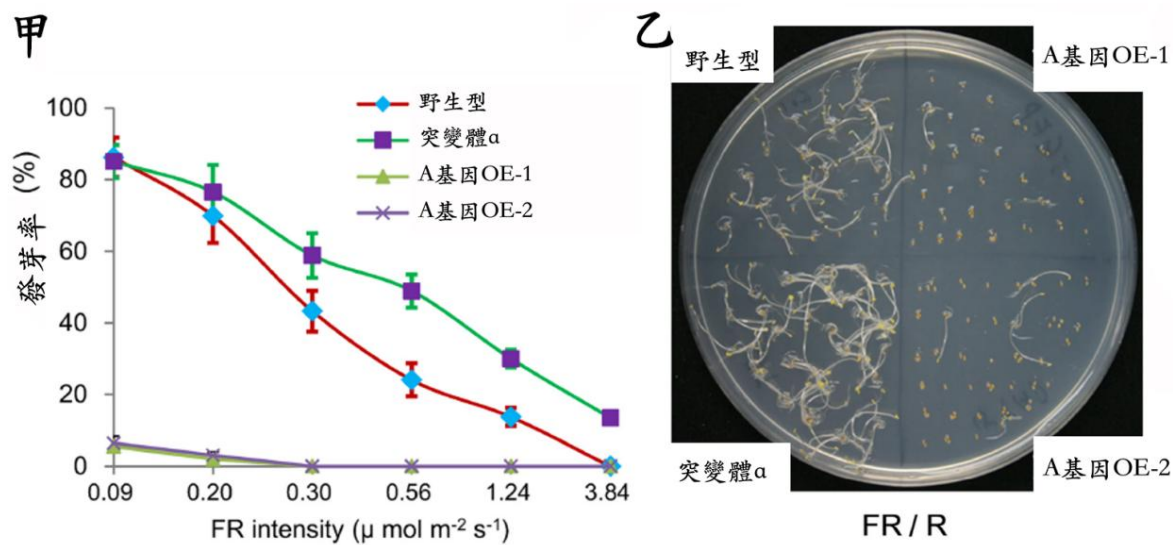
What type of structure is located at point C in diagram B? (2 points)

3. Infer the possible function of the structure at point C in diagram B. (3 points)

4. Which hormone has the most significant effect on the formation of the structure at point C in Figure B? (2 points)

5. Infer what kind of hormone transmission plant gene X might be involved in? (3 points)

V. The Influence of Light on Plant Seed Germination



A student was very interested in how light affects plant seed germination. Therefore, using wild-type Arabian mustard, mutant a, and transgenic plants with highly expressed genes (A gene OE-1 and A gene OE-2), the student tested them under different intensities of far-red light (FR) (A), and under far-red light irradiation combined with...

The seed germination experiment was conducted using red light irradiation treatment (FR/R) (B), and the results are shown in the figure above. Based on the results of this experiment, answer the following

related questions: 1. Is the regulation of Arabian mustard seed germination by gene A affected by phytochrome?

Yes; No (please check the box) (2 points)

Why? (4 points) Explain

your reasoning:

2. What role does gene A play in regulating seed germination?

Positive regulator; Negative regulator (please check the box) (1 point)

Why? (3 points) Explain

your reasoning:

2021 Republic of China Biology Olympiad National Team

Selection Camp

Theoretical Assessment Test Questions

Part Three

Student Codey



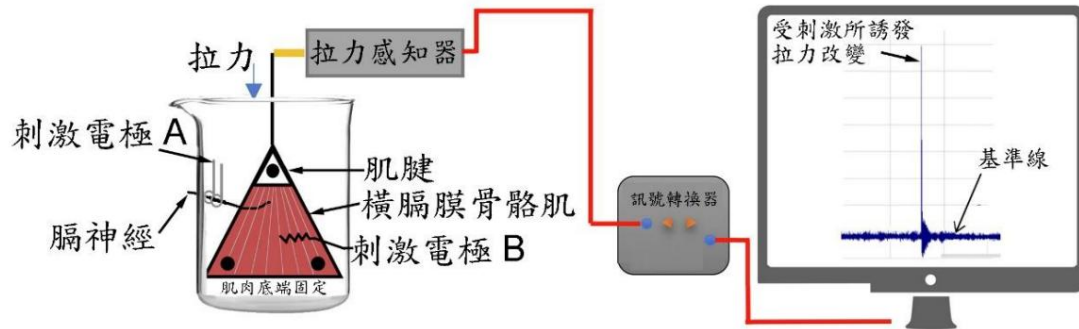
Precautions:

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y

Part Three: (100 points)

- I. The diaphragm and phrenic nerve of mice are often isolated for screening the mechanisms of action of certain toxic substances. As shown in the figure, stimulation electrodes can be arranged to induce action potentials in the phrenic nerve (stimulation electrode A) or to induce diaphragmatic muscle contraction (stimulation electrode B).



- (1) Please describe the mechanism by which the diaphragm contracts after the action potential of the phrenic nerve is induced by stimulating electrode A. 0.5 points for each key point, maximum 5 points.

- (2) A graduate student was able to record muscle contraction when electrode B was activated in this experiment, but no muscle contraction was recorded when electrode A was activated. What error might have occurred in the solution preparation? (5 points)

It is known that poison X can induce respiratory failure and death in rats. A graduate student used the diaphragm and phrenic nerve group of rats...

The study investigated the mechanism of action of poison X and obtained the following results:

圖一

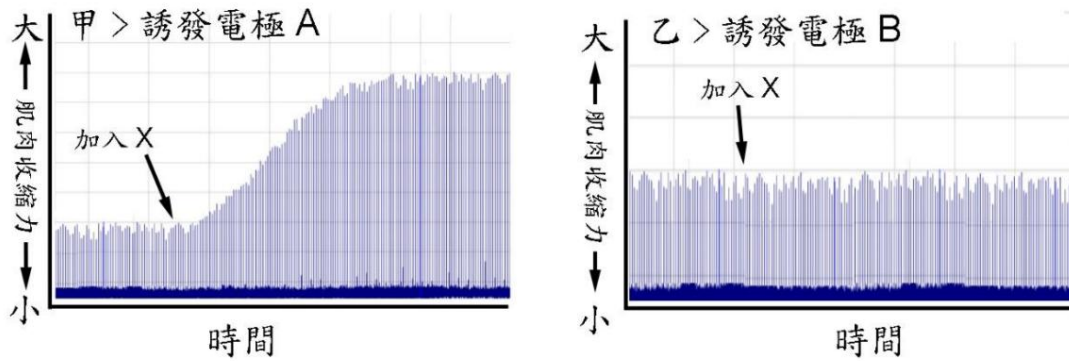


Figure 1: Assessing changes in muscle contractility using electrical stimulation. Each upward line in the figure represents a change in muscle contractility.

The magnitude of muscle contraction force after secondary stimulation

圖二

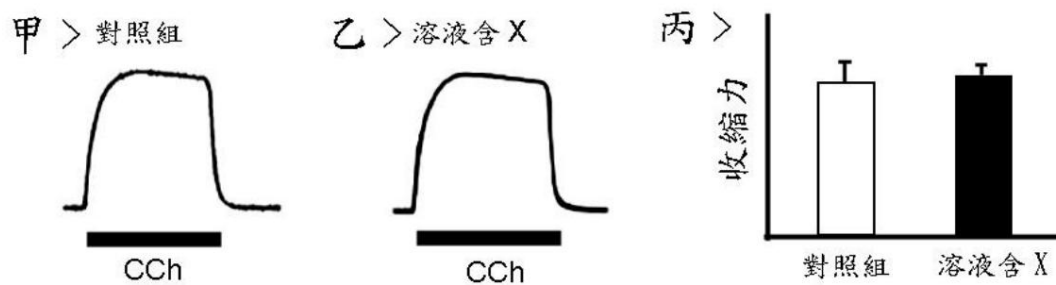


Figure 2: Comparison of the effects of added CCh (a substance that can mimic the action of acetylcholine) on muscle function.

When a chemical substance (such as a substance that can be relatively stable in water and is not destroyed by enzymes) is placed in an organ chamber (a container for tissues), the solution in the organ chamber contains (B) and does not contain (A). When the muscle contains X...

Changes in muscle contractility. Graph C shows the statistical results of five separate experiments.

	Organ chamber normal solution; Organ chamber solution containing X	
Non-irritating	0.03±0.02	0.04±0.02
Start Electrode A	1.35±0.18	0.29±0.25
Start electrode B	0.05±0.03	0.04±0.03

The table above shows that when the electrochemical method was used to analyze the acetylcholine content in the synaptic region, no further stimulation was initiated.

Difference in acetylcholine content measured between stimulation with only electrode A activated and stimulation with only electrode B activated.

(3) Based on the above three sets of experiments, please determine the possible modes of action of X and write down the reasons (15 points).

II. Please correctly match the following options with slugs and leeches (15 points) (One point will be deducted for each incorrect option, up to a total of 15 points)

A. Body segmentation; G. Abdominal suckers; M. Movement via abdominal feet.	S has male pores	Y no eyespot
B. Body unsegmented; H. Has a mouth sucker; N. Moves using suckers.	T has a female opening	Z feeds on blood.
C. No anus; I. Has abdominal feet; O. Extends and contracts body during movement; U. Has a seminal vesicle opening.		
D. Anus at the tail. J. Tentacles present. P. Body does not extend or retract during movement. V. Only one genital opening.		
E. Anus on the right side; K. No tentacles; Q. Has a breathing hole.	W has a pair of eye spots	
F. Has a tail sucker; L. Has mucus on its body surface; R. Has no respiratory pores.	X has multiple eye points	

Slug:

Leeches:

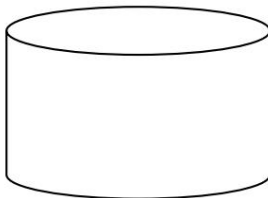
III. Genuine dried scallops come from **(1)** a specific group within the phylum Animalia (1 point) **and (2)** a class (1 point).

(3) (Name of part) (1 point), but there are always unscrupulous vendors who use fish paste for molding, cut stingray fins into pieces, or

They used pieces of squid legs and body to deceive customers. They were only allowed to use the knives, forks, and chopsticks provided at the table.

In the case of toothpicks, (4) please briefly explain how to distinguish whether the cooked dried scallops in front of you are genuine? (7

point)



Simple illustration of dried scallops

(1) _____ Phylum Animalia

(2) _____ Outline

(3) _____ (Part Name)

(4) Please briefly explain how to distinguish whether the cooked dried scallops in front of you are genuine.

IV. Progesterone (P4) is a very important hormone in human reproductive physiology, especially during pregnancy.

Regarding the roles of progesterone in early conception, implantation, and maintaining a normal pregnancy, which of the following statements are correct or incorrect?

(1 point per square, 5 points in total) Please mark with a checkmark (y).

	Correct or incorrect	
(A) Promotes smooth muscle contraction and ciliary activity in the fallopian tube.	X	
(B) Induces cervical mucus secretion	X	
(C) Inhibit prolactin-induced lactation	X	
(D) Raise body temperature	x	
(E) Produces negative feedback inhibition on the hypothalamus and pituitary gland	x	

V. After fertilization, a zygote is formed. At the time of embryo implantation, the uterine lining is in the secretory phase, and blood supply is reduced after implantation.

The endometrium should become more abundant, and at this stage, it thickens further; this endometrium is called the decidua.

These changes are called the decidual response. Based on the positional relationship between the decidua and the embryo, it is divided into three parts: y basal decidua...

The uterine lining consists of three layers: y the decidua basalis (the innermost layer of the uterus); y the decidua capsularis (the outermost layer of the uterus), which covers the embryo; and y the decidua parietalis (the outermost layer of the uterus). Progesterone plays a crucial role in this process.

color.

Preterm birth (PTB) mainly refers to delivery before 37 weeks of gestation. Studies show that the incidence of premature birth is higher each year.

Globally, approximately 15 million women are born prematurely, with a prevalence rate of about 5% to 18%. In 2018, the incidence of premature births in the United States...

The rate is 10%, which is also a major cause of neonatal death or serious sequelae in perinatal infants. Due to lung and...

The nervous system develops fully only in the late embryonic stage, therefore premature birth is also associated with childhood lung diseases and neurological disorders.

Developmental disorders are a major cause of illness, and according to statistics, related medical costs amount to as much as \$26 billion annually.

The study indicates that 80% of postpartum thrombosis (PTB) is spontaneous, while 20% is caused by a deteriorating maternal or fetal condition.

This is caused by various factors. Generally speaking, although more than half of premature births are due to upper genital tract infections, placental abruption, and...

Multiple births can cause this, but there is also evidence suggesting that the mother may have depression or post-traumatic stress disorder.

The resulting psychological stress can also lead to abnormal embryo implantation and is closely related to primary preterm birth.

The molecular mechanism remains unclear.

FKBP51 is a protein that is highly active under stress and is currently believed to be associated with depression or stress.

It is related to mental illnesses caused by stress; and these mental illnesses caused by depression or stress are also related to PTB.

Recent research indicates that transgenic mice that cannot express the FKBP51 protein can prevent stress-induced...

The hormonal changes and depressive and anxious behaviors that occur indicate that FKBP51 is associated with stress-related illnesses.

Closely related (Guzeloglu-Kayisli et al., PNAS, 2021; 118(11): e2010282118). Please refer to...

Based on the above explanation, answer the following questions:

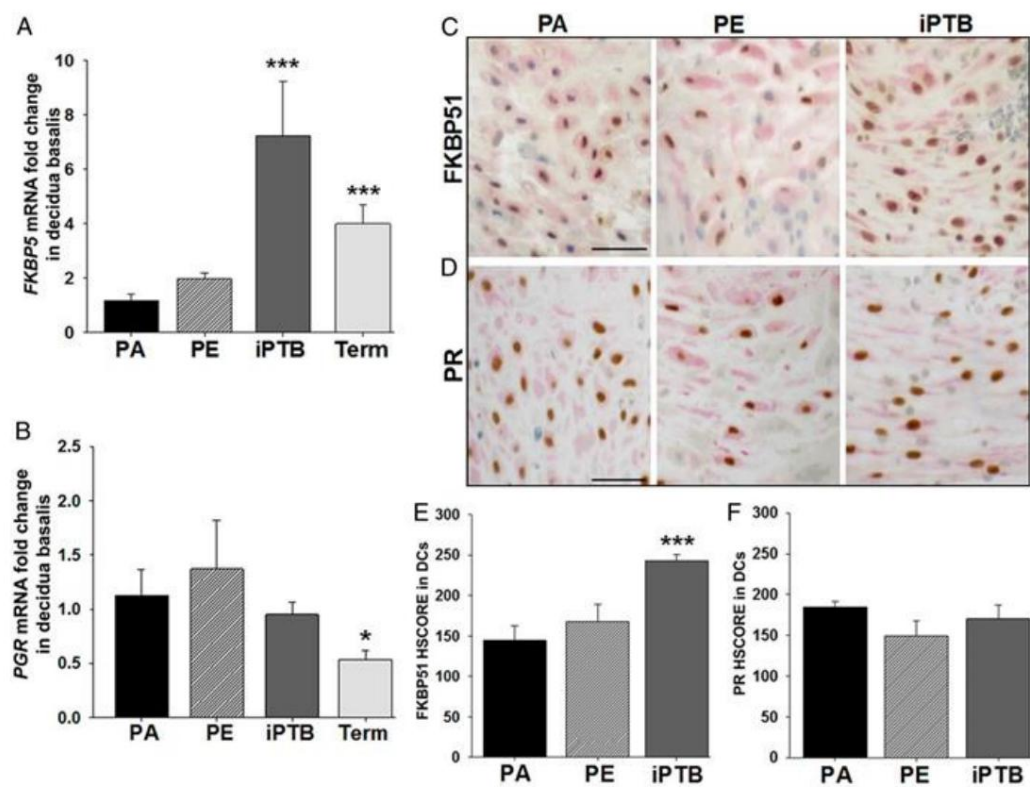


Illustration:

FKBP5 mRNA fold change in decidua basalis: (This refers to a change in the fold expression of FKBP5 mRNA at the decidua basalis.)

PGR mRNA fold change in decidua basalis: fold change in the expression of progesterone receptor mRNA at the decidua basalis.

PA (placenta accrete): Placenta accreta

PE (preeclampsia): preeclampsia; IPTB (idiopathy preterm birth): primary preterm birth.

Term: Full-term women giving birth

Figures C and D show FKBP5 and lutein receptors, respectively, using immunostaining, with brown ink.

FKBP5 HScore in DCs: Performance score of FKBP51 at the decidua basalis.

PR HScore in DCs: Performance score of luteinizing receptors at the decidua basalis.

- (1)** Please explain which group is the positive control and which group is the negative control in this experiment. (1 point for each question, 3 points in total)

	Positive Control or Negative Control
PA (placenta accrete): Placenta accreta	
PE (preeclampsia): Preeclampsia	
Term: Full-term women giving birth	

- (2)** Please describe the six sub-figures A, B, C, D, E, and F above, and summarize the results shown by these six sub-figures in one sentence (in Chinese, within 15 characters; one English noun is counted as one character, and one point will be deducted for each extra character) (9 points)

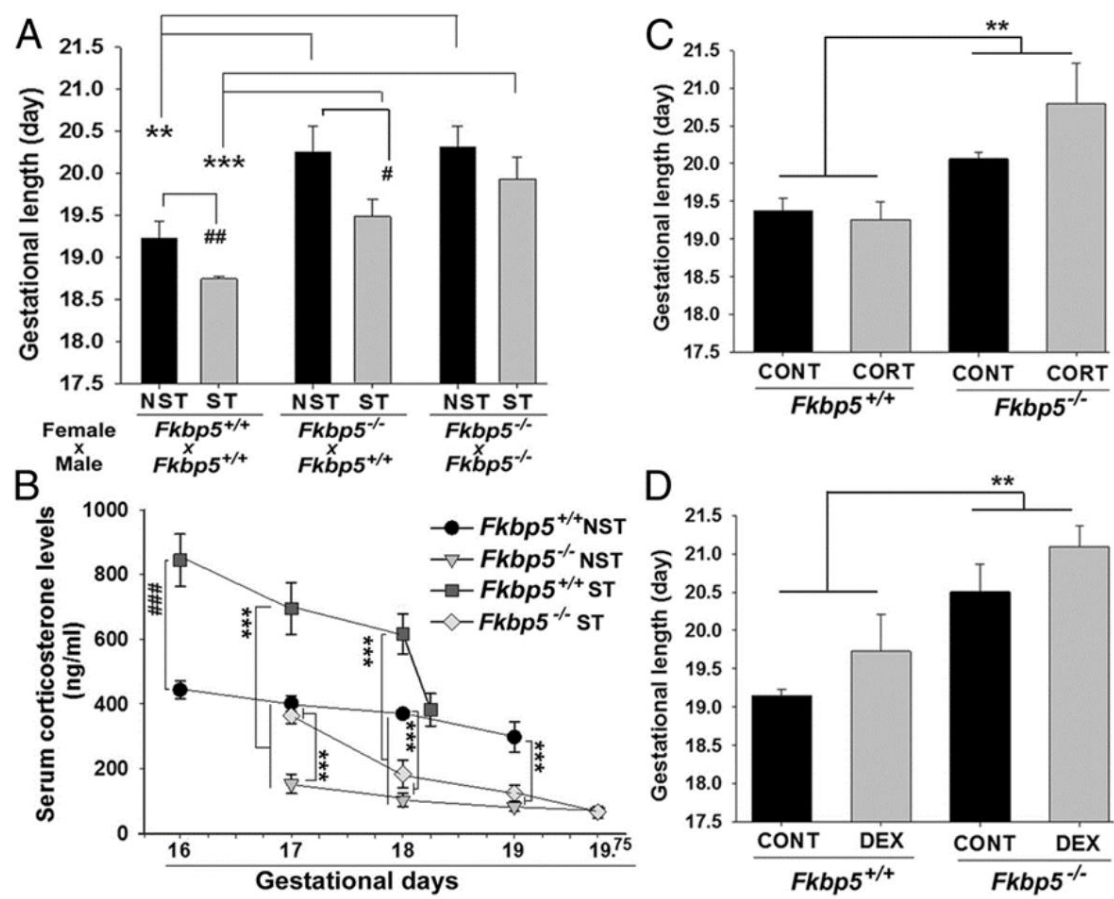


Image caption:

Gestational length (days): Pregnancy duration (days); NST: No emotional stress; ST: Emotionally stressful.

Force. Female: Female mouse; male: Male mouse; *Fkbp5*^{+/+}: *Fkbp5* gene knockout mouse; *Fkbp5*^{+/+}: Wild type; serum corticosterone levels (ng/ml): Corticosterone content in blood (ng/ml); CONT: Control Preparation group; CORT: Cortisterone (10 mg/kg) added; DEX: Dexamethasone (200 μ g/kg) added.

(3) Please explain the four diagrams A, B, C, and D respectively. (4 points)

(4) Please summarize the meaning of the four diagrams ABCD in one sentence (in Chinese, within 15 characters; each English noun is counted as one character, and one point will be deducted for each extra character). (4 points)

six,

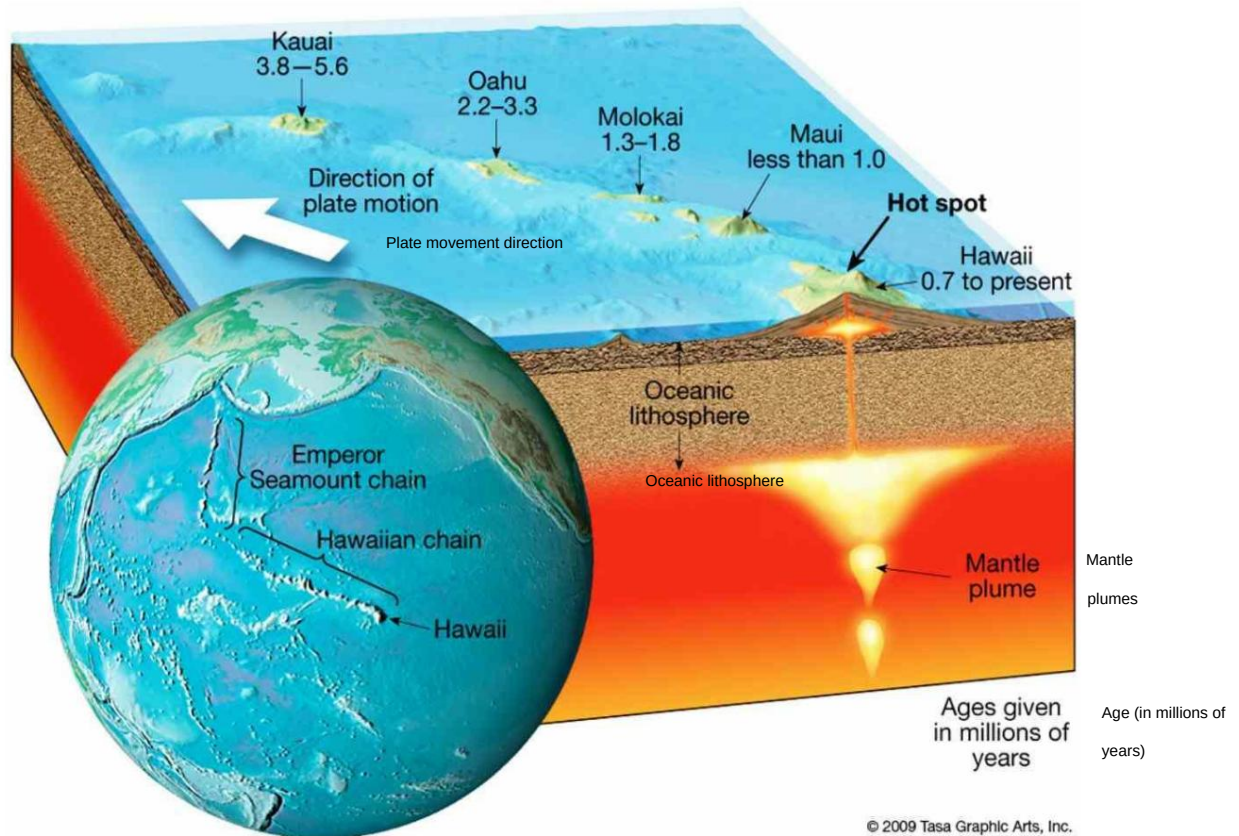


Image: A diagram showing the order and chronology of the formation of the islands in the Hawaiian Islands.

Beginning in 1991, the wild field crickets (*Teleogryllus oceanicus*) on *Kauai Island, Hawaii*, were attacked by the parasitic fly *Ormia ochracea*, which invasive from North America. This parasitic fly specifically targets crickets that can make sounds to attract mates. Because only male crickets rub their wings together to make sounds, the parasitic behavior of *Ormia ochracea* caused a sharp decline in male crickets, and by 2001, crickets had become very rare on Kauai Island.

However, in 2003, researchers at the University of California, Riverside, discovered that the cricket population on Kauai Island had rebounded, but a staggering 96% of the male crickets were silent. Apparently, on Kauai Island, male crickets avoided being parasitized by the Omega brown fly by not making a sound; these silent male crickets were termed "flatwings" by the researchers.

In 2005, researchers at the University of St Andrews in the UK found that 45% of the male crickets on Oahu, about 100 kilometers east of Kauai, had also become silent; however, on the Big Island of Hawaii, the easternmost island of the archipelago, only 2% of the male crickets were silent.

Based on the findings of the above survey, please answer the following questions:

(1) To which order of insects does the cricket belong? Answer: _____ (2 points)

(2) The sounding pattern of crickets is: (A) wing-wing sounding; (B) wing-leg sounding; (C) leg-leg sounding; (D) wing-belly sounding?

Answer: _____ (3 points)

- (3) If scientists wanted to use single nucleotide polymorphism (SNP) analysis to understand whether the non-vocalization behavior of crickets on Kauai and Oahu islands evolved independently, under what SNP analysis results would you confirm whether the non-vocalization behavior of the crickets on the two islands evolved independently or not independently? (10 points)
- (4) If you were to design an experiment to understand the relationship between the timing of the parasitic fly invasion and the prevalence of the silent trait among crickets on the Hawaiian Islands, what kind of data would you collect to test your hypothesis? (10 points)

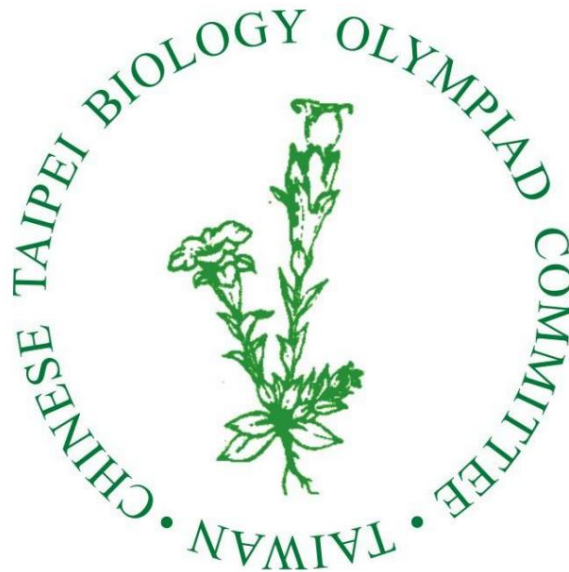
2021 Republic of China Biology Olympiad

National Team Selection Camp

Theoretical Assessment Test Questions

Part Four

Student Code: Preparatory Paper



Precautions:

1. This section of the test consists of 5 pages (including the cover) and is worth 100 points.
2. Please check for missing pages. If any pages are missing, please inform the test administrator immediately.
3. When answering the questions, please make sure that the cover of the exam paper matches your exam number.
4. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.
5. You may write your answers on the back of the questions, but please indicate the question number.

Part Four: **(100 points)**

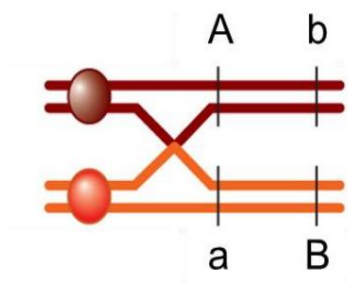
1. Using Mendel's hybridization experiments of pure purple-flowered pea lines and pure white-flowered pea lines, please present the results, including the parent generation, the first progeny, and the second progeny.

From the two offspring, the "law of segregation" is derived. (25 points)

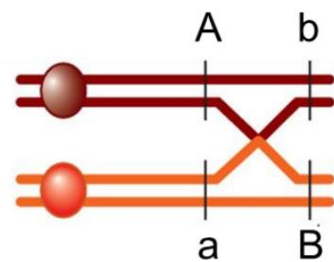
2. Explain why Morgan's white-eyed fruit fly experiments support the hypothesis that "genes are located on chromosomes." (25 points)

3. Please provide examples to illustrate "maternal inheritance" and "maternal effect ." (25 points)

4. During meiosis in germ cells, homologous chromosomes pair and exchange. Similarly, during mitosis in somatic cells, after chromosome replication, homologous chromosomes occasionally pair and exchange, sometimes producing daughter cells with different phenotypes from the mother cell. This phenomenon is called mitotic crossing over. If genes A and B are located on the same chromosome, and allele A is dominant over allele a, and allele B is dominant over allele b, then an individual with the phenotype (A dominant, B dominant) and the genotype $AaBb$ might produce daughter cells with both (A recessive, B dominant) and (A dominant, B recessive) phenotypes if one cell in this individual undergoes mitotic crossing over as shown in Figure 1. If one cell in this individual undergoes mitotic crossing over as shown in Figure 2, then (A dominant, B recessive) daughter cells might appear, but (A recessive, B dominant) daughter cells will not appear. What type of mitotic crossing over is required to produce only (A recessive, B dominant) daughter cells, without producing (A dominant, B recessive) daughter cells? Please draw a flowchart to illustrate the process, including the location of the crossing over, the allelic composition of the chromosomes after the crossing over, the chromosome arrangement during mitosis, and the chromosome composition in the daughter cells. (25 points)



(Figure 1)



(Figure 2)

2021 Republic of China Biology Olympiad National Team

Selection Camp

Theoretical Assessment Test Questions

Part Five

Student Code: Preparatory Paper



Precautions:

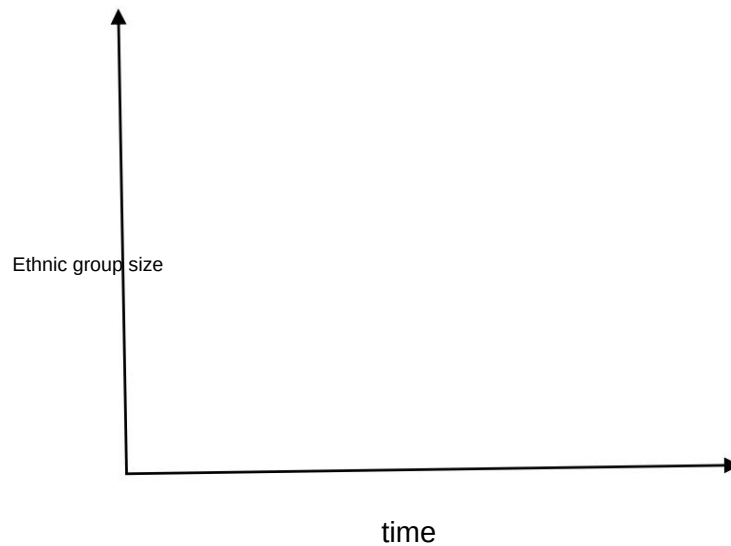
1. This section of the test consists of 11 pages (including the cover) and is worth 100 points.
2. Please check for missing pages. If any pages are missing, please inform the test administrator immediately.
3. When answering the questions, please make sure that the cover of the exam paper matches your exam number.
4. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.
5. You may write your answers on the back of the questions, but please indicate the question number.

Part Five: (100 points)

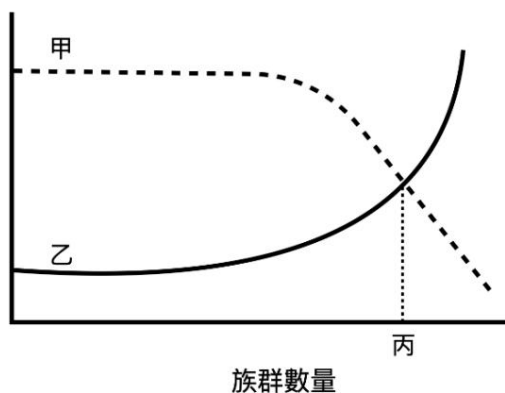
- I. The growth curve of biological population size includes at least exponential growth (also known as J-type growth) and logistic growth (also known as S-type growth or logical growth). Among them, the population size of logistic growth will be limited to a population size called carrying capacity (K) due to competition.

1. Represent the exponential and logarithmic growth curves, which have the same intrinsic rate of increase (r).

Draw the line (including the one marked K) in the diagram below and label it clearly. (Total 8 points)



2. A closed population (without immigration or emigration) is affected by its birth rate and death rate. If a population exhibits logarithmic growth and has a carrying capacity of K, the relationship between its birth rate, death rate, and population size can be represented by the following graph. What parameters do curves A and B in Figure 1 represent? What is the population size of curve C? (6 points total)



(Figure 1)

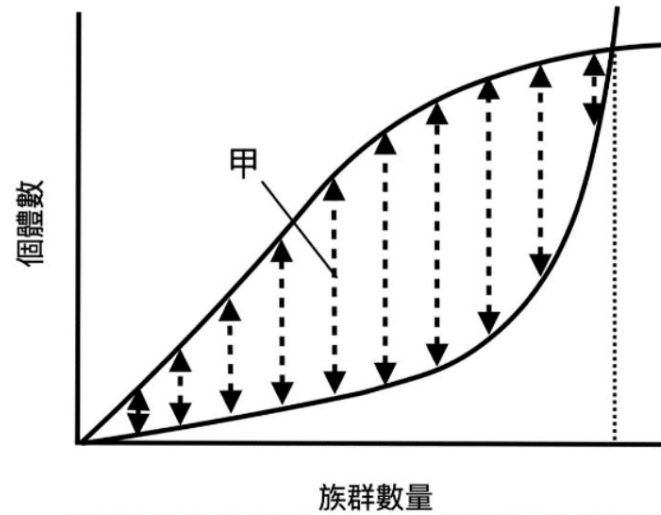
First:

Second:

C:

3. Continuing from the previous question, if the birth rate and death rate curves are replaced with the number of births and deaths respectively, then Figure 1 can be modified.

Present it as shown in Figure 2. What does A in Figure 2 represent? (3 points)



(Figure 2)

Answer:

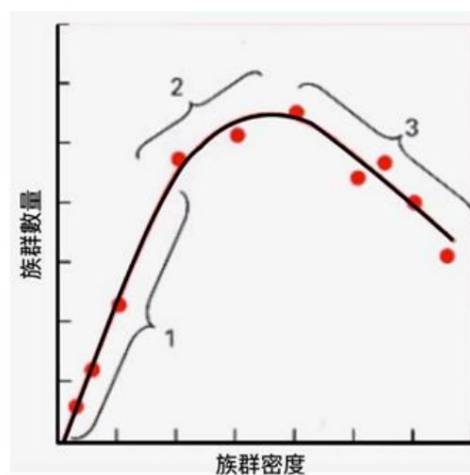
4. The survival rate of individuals in a logarithmic growth population is, to some extent, controlled by population density, a phenomenon known as density-dependent. If the relationship between population size and population density of an organism is observed as shown in Figure 3, compare the degree to which density dependence affects population size changes in each of segments 1-3, and select the most correct option: (3 points)

(A) $1 = 2 = 3$

(B) $1 > 2 > 3$

(C) $3 > 2 > 1$

(D) $3 > 2 = 1$



(Figure 3)

5. The growth pattern of a continuously reproducing population can be simply represented by a mathematical model. For example, exponential growth.

The growing population:

$$\frac{dN}{dt} =$$

Where N is the population size, dN/dt represents the population change per unit time, and r is called the internal growth rate.

(intrinsic rate of increase), when $r > 0$, and the higher the value, the faster the population grows.

Conversely, a population exhibiting logarithmic growth and a loading of K can be represented by the following mathematical model:

$$\frac{dN}{dt} = \left(\frac{-B}{\text{First}} \right)$$

The numbers of ethnic groups A and B are respectively: (4 points in total)

First:

Second:

6. Following on from the previous question, a species usually does not exist alone; when it coexists with other species, competition occurs.

The population size of a species is influenced not only by competition among individuals of the same species, but also by competition with other individuals of the same species.

The species are affected by competition. Assuming two competing species exist, how will their populations grow?

The following mathematical models can be used to represent the population sizes of the two species (N_1 and N_2 represent the population sizes of the two species, and r_1 and r_2 represent the population sizes of the two species).

The intrinsic growth rate of a species, where γ_{12} represents the negative impact of species 2 on a single individual of species 1, and γ_{21} represents the negative impact of species 1 on the species.

The degree of negative impact of 2 individual species:

$$\frac{dN_1}{dt} = \left(\frac{\gamma_{B1}(\gamma_C)}{\text{First}} \right)$$

$$\frac{dN_2}{dt} = \left(\frac{\gamma_{12}(\gamma_{21})}{\text{Man}} \right)$$

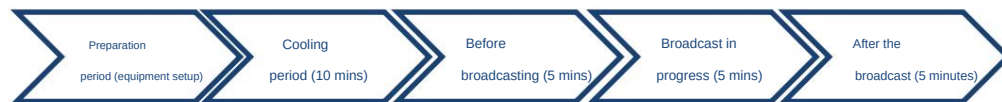
The number of ethnic groups A through F are respectively: (Total 9 points)

First: _____ ; Second: _____ C: _____

Man: _____ ; E: _____ Self: _____

II. To explore the influence of vocalization behavior among territorially vocalizing species on each other, researchers conducted three indoor experiments, playing different processed cicada sounds to wild Light-vented Bulbuls to analyze the effects of different cicada species, sound frequencies, and volumes on bird singing behavior. The subjects were 19 wild Light-vented Bulbuls, all adult males, each kept in an individual cage in a aviary. The Light-vented Bulbul's song is quite varied, somewhat resembling a "chocolate" sound, with calls ranging from rounded single notes to triplets of "chirp chirp chirp." During the experiment, the average sound intensity of these Light-vented Bulbuls was 85 decibels.

The experiment was conducted in spring. The researchers moved the 19 Light-vented Bulbuls to seven quiet experimental spaces the day before the experiment, placing two to three individuals in each space. The following morning, a video recorder and audio equipment were set up in front of each space. Once the equipment was set up, the personnel left the experimental space and began recording. Ten minutes later, the audio recording began, with each recording lasting 5 minutes. Five minutes were taken before and after each recording as representative values, so a single experiment took a total of 25 minutes (Figure 1). After the experiment, the researchers reviewed the footage to count the number of calls made by each individual bulbul.



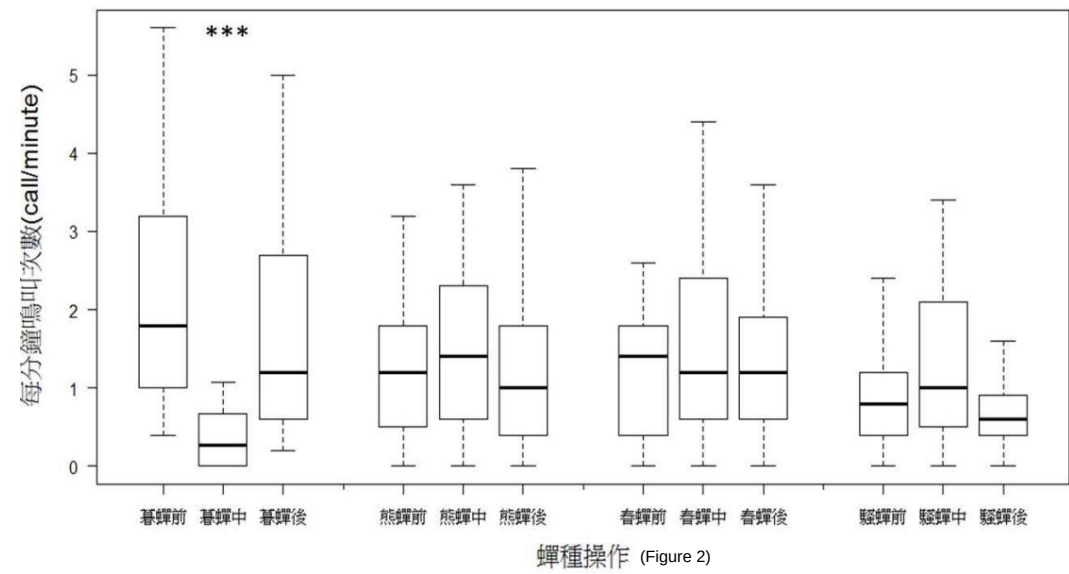
(Figure 1) Flowchart of the broadcasting experiment. The operation time for one experiment is 25 minutes.

To investigate the influence of different cicada species' calls, researchers played the calls of four of the most common cicada species at low altitudes in Taiwan (Yangmingshan cicada, Taiwan cicada, green-faced cicada, and Taiwan bear cicada) to 19 experimental individuals of *Bulbus hainanensis*. The sound intensity of each cicada species in the wild exceeded 85 decibels, but to eliminate the influence of sound intensity, the playback intensity of each cicada species was uniformly set to 85 decibels in this experiment.

Researchers also used computers to adjust the sound frequency and intensity of the cicadas in Yangmingshan. For sound frequency, they played combinations of three different sound frequencies from the cicadas (original frequency, all reduced by 500 Hz, all increased by 500 Hz). The combination of reduced 500 Hz had the highest overlap with the call frequency of the bulbul (80.5%), followed by the original frequency (70.9%), and the combination of increased 500 Hz had the lowest overlap (59.5%). For sound intensity, they played three different sound intensities from the cicadas (80 dB, 85 dB, and 90 dB). All 19 bulbul experimental individuals underwent testing with four cicada species, three sound frequencies, and three sound intensities.

The experiment was played eight times, with at least one day between each play.

The experimental results are presented in the following three figures.



(Figure 2) Broadcasting of the calls of different cicada species (Yangmingshan cicada, Taiwan bear cicada, green-faced spring cicada, and Taiwan slug cicada).

The number of times the white-headed bulbul chirps per minute is controlled. The white-headed bulbul only chirps when playing the Yangmingshan evening cicada song.

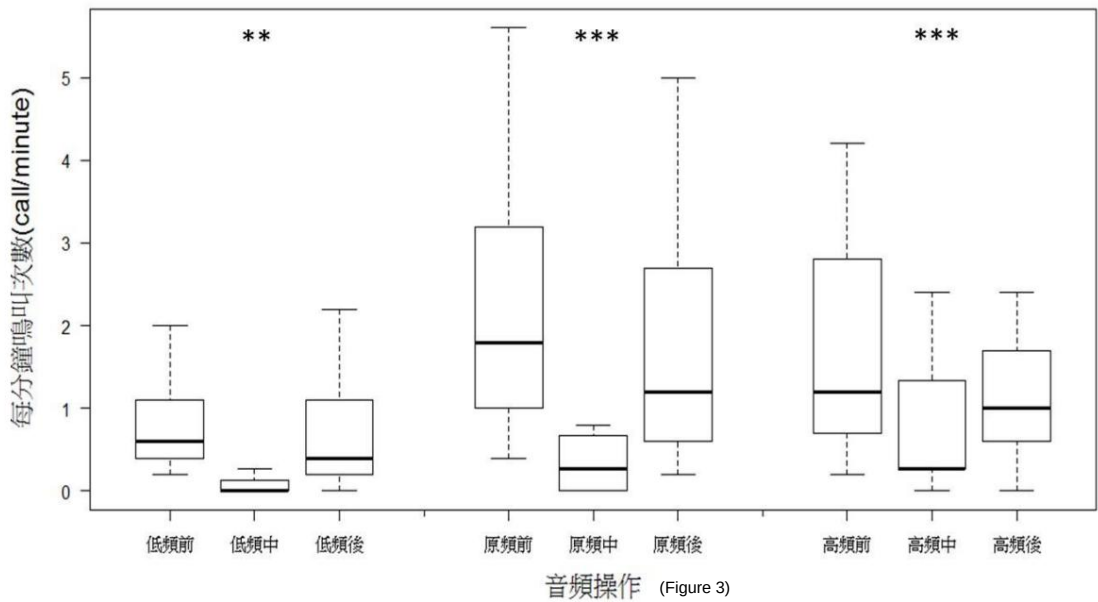
The number of chirping events during broadcasting was significantly lower than before broadcasting ($p < 0.001$), and the number of chirping events during broadcasting was significantly lower than after broadcasting ($p < 0.001$). There was no significant difference in the number of chirping events before and after broadcasting ($p = 0.941$). The other three cicada species...

During the broadcasting operation, the number of calls of *Bulbus paedonia* did not differ significantly among the three broadcasting periods (*Cicada taiwanensis* group: $p = 0.587$,

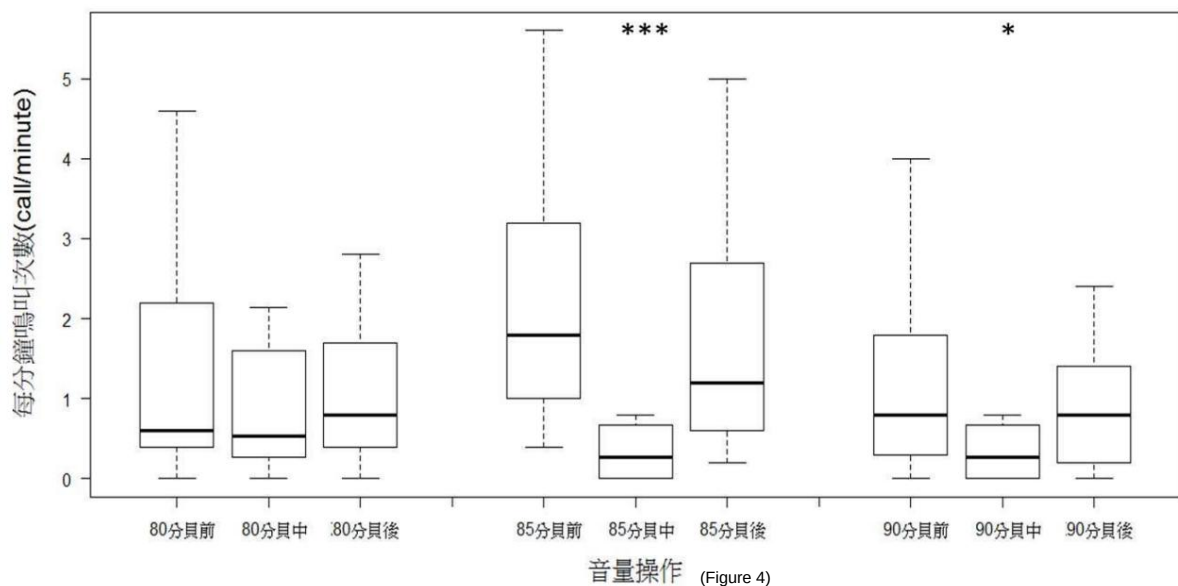
Cicada simonii group: $p = 0.458$, *Cicada taiwanensis* group: $p = 0.053$). *Bulbus paedonia* was present in four different cicada species.

The number of calls during the vocalization broadcast was significantly lower in the Yangmingshan cicada group than in the Taiwan bear cicada group ($p = 0.035$).

The *Shanmu* cicada group was significantly lower than the *Qingmian* cicada group ($p = 0.032$), while there were no significant differences between the other pairs.



(Figure 3) Number of chirps per minute of the White-headed Bulbul during broadcasts of different frequencies of Yangmingshan cicada sounds. When playing low-frequency (500 Hz lower) Yangmingshan cicada sounds, the number of chirps by the White-headed Bulbul during the broadcast was significantly lower than before the broadcast ($p = 0.003$), and the number of chirps during the broadcast was significantly lower than after the broadcast ($p = 0.020$). There was no significant difference in the number of chirps before and after the broadcast ($p = 0.773$). When playing the original frequency Yangmingshan cicada sounds, the number of chirps by the White-headed Bulbul during the broadcast was significantly lower than before the broadcast ($p < 0.001$), and the number of chirps during the broadcast was significantly lower than after the broadcast ($p < 0.001$). There was no significant difference in the number of chirps before and after the broadcast ($p = 0.941$). When playing high-frequency (increased by 500 Hz) cicada chirping sounds from Yangmingshan, the number of chirpings by the white-headed bulbul during the broadcast was significantly lower than before the broadcast ($p < 0.001$), and the number of chirpings during the broadcast was significantly lower than after the broadcast ($p = 0.008$). There was no significant difference in the number of chirpings before and after the broadcast ($p = 0.449$). There was no significant difference in the number of chirpings during the middle of the broadcast across the three frequency broadcasting operations ($p = 0.063$).



(Figure 4) Number of chirps per minute of the White-headed Bulbul under different sound intensities of Yangmingshan cicada broadcasts. When the Yangmingshan cicada sound at a volume of 80 decibels was played, there was no significant difference in the number of chirps of the White-headed Bulbuls before, during, and after the broadcast ($p = 0.054$). When the Yangmingshan cicada sound at a volume of 85 decibels was played, the number of chirps of the White-headed Bulbuls during the broadcast was significantly lower than before the broadcast ($p < 0.001$), the number of chirps during the broadcast was significantly lower than after the broadcast ($p < 0.001$), and there was no significant difference in the number of chirps before and after the broadcast ($p = 0.941$). When the sound of cicadas chirping at dusk in Yangmingshan was played at a volume of 90 decibels, the number of chirps by the white-headed bulbul during the broadcast was significantly lower than before the broadcast ($p = 0.016$). There was no significant difference in the number of chirps during and after the broadcast ($p = 0.226$), and there was no significant difference in the number of chirps before and after the broadcast ($p = 0.441$). There was no significant difference in the number of chirps during the middle of the broadcast across the three broadcast frequencies ($p = 0.389$).

Which of the following statements is true? Write "O" before the question number if it is true, and "X" if it is false. (33 points, 5 points deducted for each incorrect answer)

- _____ 1. The calls of different cicada species can influence the singing behavior of the white-headed bulbul. The white-headed bulbul hears the evening cicadas of Yangmingshan...

When the Taiwan bear cicada, the green-faced cicada, and the Taiwan shrew cicada are singing, they all reduce their singing behavior.

- _____ 2. The intensity of the cicadas' calls at dusk on Yangmingshan Mountain may be higher than that of the other three cicada species, thus causing [the noise/sound] to [cause] [the noise/sound] in the local area.

The study found that when white-headed bulbuls hear the chirping of cicadas at dusk in Yangmingshan, they reduce their chirping behavior.

- _____ 3. The frequency overlap between the Yangmingshan cicada's evening song and the white-headed bulbul's song may be higher than that between the other three cicada species and

the white-headed bulbul. Therefore, in this experiment, the white-headed bulbul reduced its singing behavior when it heard the Yangmingshan cicada's evening song.

- _____ 4. The time, space, and audio range of sound transmission are limited resources for organisms that communicate using sound signals. To reduce competition for sound, vocal species

sharing a common domain often differentiate in the use of audio range or the timing of vocalization.

- _____ 5. There is vocal competition between the White-headed Bulbul and the Yangmingshan Night Cicada, and the White-headed Bulbul has a relatively stable wild environment in Taiwan.

A competitor with a disadvantageous voice.

- _____ 6. Among the four cicada species, the Yangmingshan evening cicada is a relatively strong competitor in terms of sound.

- _____ 7. Based on the results of this experiment, the frequency of the cicada's chirping at dusk in Yangmingshan does not affect the singing behavior of the white-headed bulbul.

- _____ 8. Based on the results of this experiment, the sound intensity of the cicadas chirping in Yangmingshan does not affect the singing behavior of the white-headed bulbuls.

- _____ 9. Based on the results of this experiment, the cicada group in Figure 2, the original frequency group in Figure 3, and the 85 dB group in Figure 3 are all relevant.

The number of times the white-headed bulbuls called all came from the same set of experimental data.

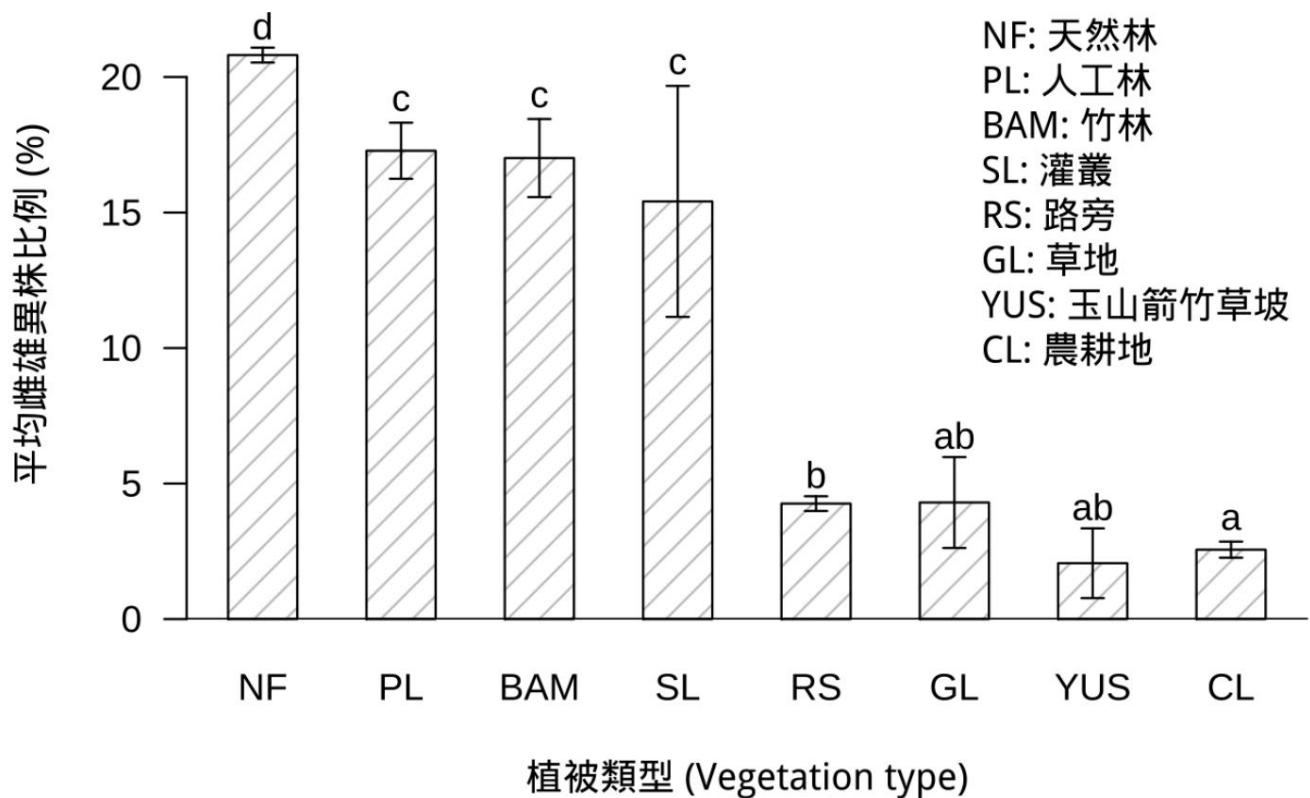
- _____ 10. The singing behavior of the White-headed Bulbul is mainly for courtship and territorial marking. In the wild in Taiwan, the White-headed Bulbul...

They will avoid the cicada's chirping season during the breeding season.

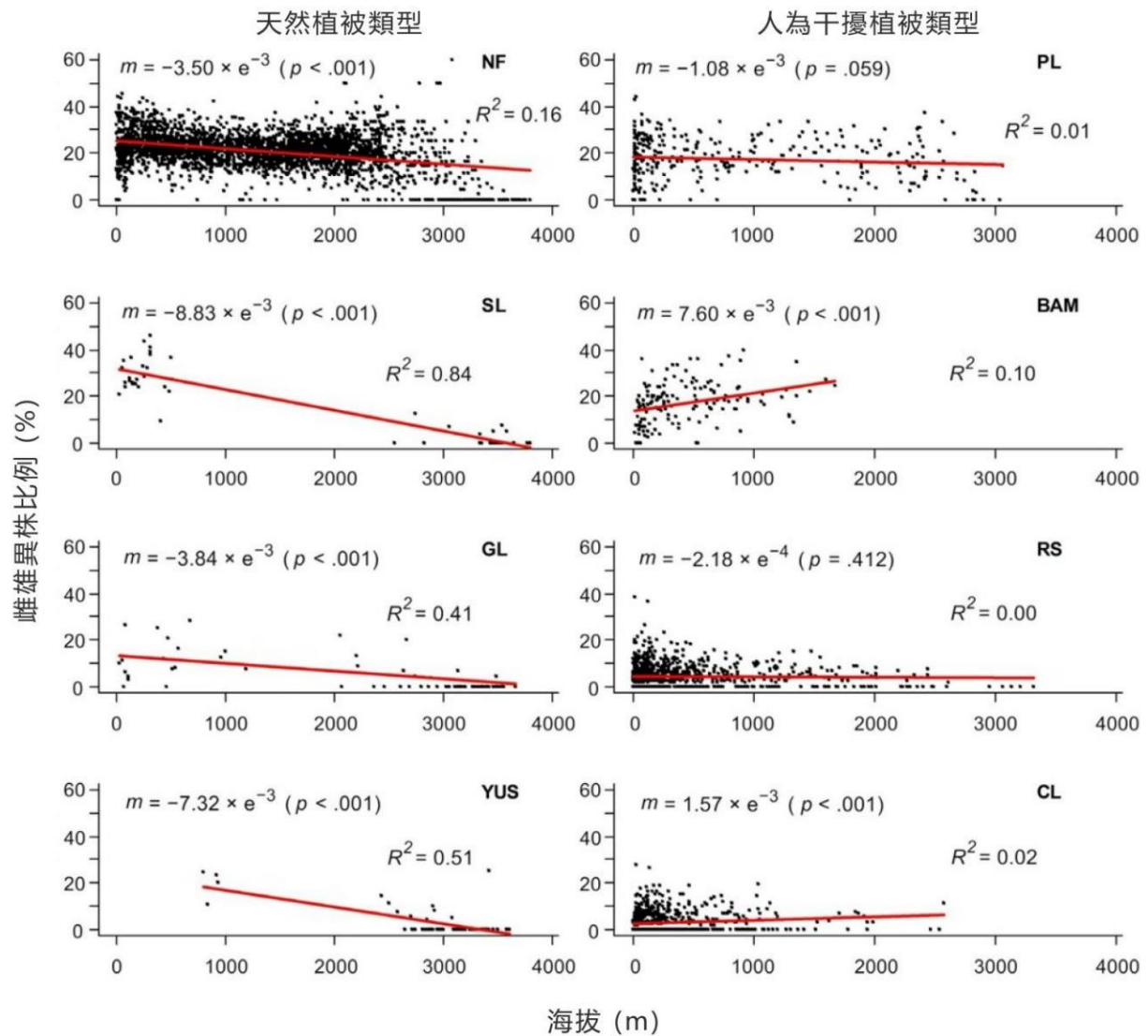
- _____ 11. The 10-minute cool-down period in this experiment was likely to minimize the impact on the behavior of the experimenters.

ring.

III. Dioecious plants are relatively rare in ecosystems. These plants mainly rely on insects for pollination. This reproductive type can increase gene exchange between individuals and avoid the negative effects of inbreeding. Past studies have also found that the proportion of dioecious plants varies in different regions, which may be related to environmental differences and biota in different regions. To understand the relationship between environmental changes and vegetation type in relation to the proportion of dioecious plants in different regional vegetation types, researchers analyzed survey data on natural vegetation and introduced plant species in Taiwan. The researchers divided the data into eight different vegetation types (four of which were natural vegetation and the other four were vegetation types affected by human disturbance), compared the proportion of dioecious plants in different vegetation types (Figure 1), and further analyzed how the proportion of dioecious plants changed with increasing altitude in these eight vegetation types (Figure 2) (Image modified from Lin et al. 2020 Ecological Research). Please refer to Figures 1 and 2 to answer the following questions:



(Figure 1) The average proportion of dioecious plants in 8 vegetation types in Taiwan. The error bars in the bar chart represent the standard error of 2 units. The letters in the figure represent the results of the Tukey HSD test. Different letters indicate that the means are significantly different.



(Figure 2) The proportion of dioecious plants in eight vegetation types in Taiwan varies with altitude. The horizontal axis represents altitude, and the vertical axis represents the proportion of dioecious plants (%). The formulas in the figure represent the results of regression analysis. R^2 is the coefficient of determination for the regression analysis, ranging from 0 to 1. A higher value indicates a stronger correlation between altitude and the proportion of dioecious plants. The p-value represents the significance of the model; when $p < 0.05$, it indicates a significant difference between the slope (m) estimated by the model and 0. NF: Natural forest; SL: Shrubland; GL: Grassland; YUS: Yushan arrow bamboo grass slope; PL: Plantation forest; BAM: Bamboo forest; RS: Roadside; CL: Farmland.

Multiple choice questions: (3 points each, 18 points in total)

- _____ 1. In which of the following vegetation types is the plant composition with the highest ratio of dioecious plants?
(A) Planted forest. (B) Natural forest. (C) Bamboo forest. (D) Shrubland.
- _____ 2. In which of the following locations is the plant composition likely to have the lowest ratio of dioecious plants?
(A) Taipei Hapen Nature Reserve. (B) Nantou Xitou Experimental Forest.
(C) Hehuan North Peak, Nantou. (D) Kaohsiung Shoushan National Nature Park.
- _____ 3. Among natural vegetation types, which of the following altitudinal ranges has the highest ratio of dioecious plants?
(A) 0–500 mý(B) 1000–1500 mý(C) 2000–2500 mý(D) 3000–3500 mý
- _____ 4. Among the following plant community types, which one has the highest ratio of dioecious plants?
(A) Yushan Juniper shrubland. (B) Juniper forest. (C) Banyan and Phoebe forest. (D) Hemlock forest.
- _____ 5. Based on this study, which of the following life forms of plants can we infer has the highest ratio of dioecious plants?
(A) Annual herb. (B) Perennial herb. (C) Shrub. (D) Tree.
- _____ 6. Based on this study, which of the following plant compositions would have the highest ratio of dioecious plants?
(A) Kenyan open forest grassland. (B) Malaysian rainforest.
(C) American Redwood Forest. (D) Siberian cold plains.

Short answer questions: (3 points each, 6 points in total)

1. Among different types of natural vegetation, which type has the highest correlation between the ratio of male and female plants and changes in altitude?
2. Among vegetation types affected by human disturbance, which type does not show a significant change in the ratio of male to female plants with altitude?

True or False: Based on this research, are the following inferences correct? Fill in "O" before the question number if correct, and "X" if incorrect. (5 points deducted for each incorrect answer, total 10 points)

- _____ 1. The average temperature of a region affects the ratio of dioecious plants in the natural vegetation of that region.
- _____ 2. Changes in vegetation type caused by human interference can lead to a significant increase in dioecious plant species.
- _____ 3. The variation trend of insect diversity with altitude gradient may be an important mechanism shaping the variation of the male-female dioecious ratio of plants in different regions of Taiwan.